ZHR at BAREC Shared Task 2026: Ordinal-Objective Ensembles with Threshold Calibration and Silver-Label Pretraining for Arabic Readability Assessment

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Abstract

We describe the ZHR systems for the two sentence-level tracks of the BAREC 2026 Shared Task on Arabic readability assessment: Strict, where training is restricted to the BAREC training split, and Open, which allows any public data. Each sentence receives one of 19 ordinal levels, and systems are ranked by quadratic weighted kappa (QWK). Both tracks use one recipe. Arabic encoders are fine-tuned under five ordinal objectives and averaged with weights from greedy selection, and the averaged score is discretized by 18 thresholds fitted to maximize QWK; every fitted quantity uses the development split. The official Strict system, an ensemble of ten train-split models, scores 85.0 QWK on the blind set. The official Open system adds pretraining on the silver-labeled BAREC-10M corpus and restricts selection to members whose early stopping never touched the calibration split; it scores 85.2. Both clear the official baseline (82.5) by 2.5 and 2.7 QWK while trailing it in exact accuracy, a trade-off that comes from the metric itself and that we measure directly. Negative results are part of the record: a catalog of 19 changes that did not help, each measured under a held-out protocol.

1 Introduction

Sentence-level readability assessment assigns each Arabic sentence one of the 19 ordinal levels of the BAREC scheme (Habash et al., 2025; Elmadani et al., 2025b). The task resists standard classification recipes. The ranked metric, quadratic weighted kappa (QWK) (Cohen, 1968), rewards getting the whole label distribution right rather than exact matches. Labels are heavily imbalanced ($140\times$ between the corpus's modal and rarest levels). In this task, the final points of QWK are decided after the forward pass, by ensembling and calibration.

We entered the two sentence-level tracks of the BAREC 2026 Shared Task (Elmadani et al., 2026a), Strict (training restricted to the BAREC training split) and Open (no data restrictions), with one system family.[1] Our official systems score 85.0 QWK on Strict and 85.2 on Open, against 82.5 for the official baseline (Table 1): $+2.5$ and $+2.7$ QWK, at lower exact accuracy, a trade-off built into the metric (§5.1).

Our contributions: (1) a documented recipe for QWK-ranked ordinal tasks (objective-diverse ensembling, greedy selection, threshold calibration), with each component's measured worth under a held-out A/B protocol and a measured ± 0.1–0.2 threshold-fit noise floor; (2) for the open setting, silver pretraining on BAREC-10M plus a contamination check that admits members by what their early stopping saw; (3) a catalog of 19 changes that did not help, reported as first-class negative results.

2 Background

2.1 Task and Data

The BAREC corpus (Elmadani et al., 2025b) provides 69,441 sentences from 1,922 documents, annotated 1–19 under published guidelines (Habash et al., 2025) and split document-disjointly into train / dev / test = 54,845 / 7,310 / 7,286; the blind set has 8,077 unlabeled sentences. The 2026 edition (Elmadani et al., 2026a) defines three tracks: *Strict* (training exclusively on the BAREC training set), *Constrained* (adds the SAMER corpus and lexicon; Alhafni et al., 2024; Al Khalil et al., 2020), and *Open* (any public data). We entered Strict and Open; the sentence-level tracks share one blind file, so a Strict system is Open-eligible.

2.2 Related Work

Readability assessment is a long-standing task; Collins-Thompson (2014) surveys the classical feature-based line and Vajjala (2022) its open prob-

[1] Strict: https://github.com/zaher-m/BAREC-Shared-Task-2026_SLR-ST
Open: https://github.com/zaher-m/BAREC-Shared-Task-2026_SLR-OT

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lems. Arabic resources run from the leveled lexicon of Al Khalil et al. (2020) and the SAMER simplification corpus (Alhafni et al., 2024) to BAREC and its guidelines (Elmadani et al., 2025b; Habash et al., 2025), with BAREC-10M (Elmadani et al., 2026b) extending the labels automatically; Liberato et al. (2024) benchmark modeling strategies across these resources and find fine-tuned encoders ahead of feature systems. In the 2025 shared task (Elmadani et al., 2025a), the top systems were transformer ensembles: the strongest strict-track system (Basem et al., 2025) applied CAMeL `d3tok` segmentation across all backbones and gained 6.3% QWK from post-processing that corrected prediction-distribution skew, evidence that on this metric what happens after the forward pass rivals the modeling. We adopt that preprocessing, and fit the discretization thresholds directly against QWK on a held-out split rather than correcting the output distribution by hand. The ordinal objectives we diversify over are established losses (Shi et al., 2023; Díaz and Marathe, 2019; de la Torre et al., 2018; Hou et al., 2016); our contribution is measuring what their diversity buys an ensemble.

3 System

Figure 1 shows the pipeline: every member is a pretrained Arabic encoder with mask-weighted mean pooling and one linear head, fine-tuned to emit a continuous readability score, and an ensemble is a weighted average of member scores, discretized once by 18 thresholds.

3.1 Members

Members differ along three axes; Figure 4 (Appendix E) reports every member's solo score.

Encoder. AraBERTv2 base and large (Antoun et al., 2020), AraELECTRA (Antoun et al., 2021), CAMeLBERT (Inoue et al., 2021), MARBERTv2 and ARBERTv2 (Abdul-Mageed et al., 2021; Elmadany et al., 2023), and AraModernBERT (Elshehy et al., 2026).

Ordinal objective. Scalar regression under mean squared error; CORN (Shi et al., 2023); soft ordinal labels (Díaz and Marathe, 2019) at $T{=}2$; cross-entropy plus a differentiable QWK penalty (de la Torre et al., 2018) at $\lambda{=}0.5$; and squared Earth Mover's Distance (EMD) (Hou et al., 2016). All emit a continuous 1–19 score; loss definitions are in Appendix A. What pays is diversity of loss geometry rather than of encoder: four of the official Strict ensemble's ten slots are AraBERTv2-base under different objectives (Figure 4).

Text variant. *Raw* applies light normalization and keeps diacritics (49.1% of training sentences are vocalized, and vocalization marks easy text); `d3tok` applies CAMeL Tools morphological segmentation (Obeid et al., 2020). On a controlled AraBERTv2-large pair, `d3tok` is worth about +6–7 QWK, more than any later modeling change; AraModernBERT is the exception and prefers raw text.

All 139 training runs (63 Strict, 76 Open) share one trainer, described in §4; per-member tags and configurations are cataloged in Appendices B and G.

3.2 Combining Members

Selection. Greedy forward selection with replacement (Caruana et al., 2004) over the member pool: each round adds the member that most improves the running ensemble's calibrated QWK on the fit split (dev in the official systems); a member's weight is the number of rounds it wins.

Thresholds. The ensemble score is discretized by 18 ordered cut points fitted with cyclic coordinate ascent to maximize QWK, worth +1–2 QWK over naive rounding. On Open, the fitted cuts are shrunk a fraction $k{=}0.35$ of the way toward the label-prior quantiles of the score distribution, which keeps sparse levels populated (algorithms: Appendix A).

3.3 Strict Track

The official Strict system is a ten-member ensemble of train-split-only models (members and weights: Appendix E), with selection, weights, and thresholds all fitted on dev; the test split is never read. Table 2 gives the build path; the official system is V6, with blind score 85.0 QWK / 38.8 Acc. We also re-ran dev-only greedy selection over all 39 train-only models to check whether a wider pool would do better: the re-selection lost to V6 on all twelve dev half-splits (-0.40 ± 0.17; §4 describes the A/B protocol), so V6 stood; greedy over a larger pool overfits its selection half.

3.4 Open Track

The Open system keeps the same pipeline and adds two ideas.

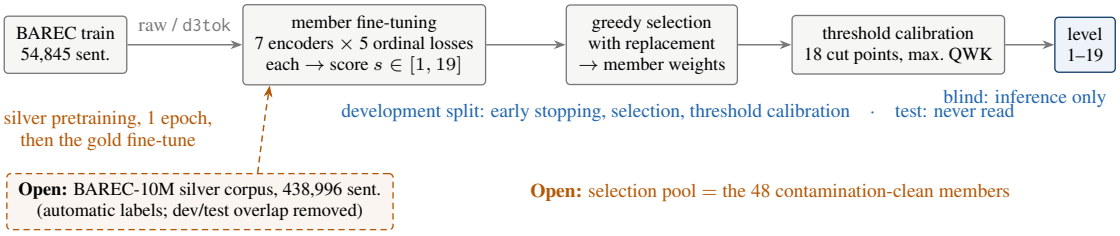


Figure 1: Both systems and what is fitted where: the shared pipeline (gray) trains on the training split, blue states the role of every other split, and orange (dashed) marks the two Open-track additions.

Silver pretraining. One epoch on 438,996 BAREC-10M sentences (Elmadani et al., 2026b) with automatic 19-level labels (8× the gold set; every row matching a dev/test sentence removed first), then the usual gold fine-tune. Against identically configured, seed-matched twins this is worth +1.00 and +0.41 solo QWK on the two matched pairs (Appendix D), the only change in either track that improved members rather than their combination. We audited the corpus before using it: its label field nearly reproduces the human gold label wherever BAREC has one (QWK 97.96 against gold), blind rows carry only model predictions, and a propensity-reweighted check found no covert advantage on the ∼57% of blind sentences it contains.

Contamination-aware selection. Members whose early stopping had touched a calibration split look better on it for free: across 18 such training runs, the best epoch on that split exceeded the mean epoch by +1.06 QWK. The official pool is therefore restricted to the 48 members whose training and early stopping never touched any calibration split (the "clean" pool). The official Open system runs greedy selection on dev over this pool; it weights 18 members, three silver models holding the largest weights (38% jointly) (membership: Appendix E). Its blind score is 85.2 QWK / 39.6 Acc. Before submission, a document-grouped dev A/B predicted it would beat the dev-calibrated V6 by +0.24 mean (11 of 12 half-splits); the realized blind gain was +0.2.

4 Experimental Setup

Split roles. Table 10 (Appendix G) states what touched which split. In the official systems, only the training split (plus, on Open, the silver corpus) receives gradient updates; the development split serves early stopping, ensemble selection, and threshold calibration, the roles the task permits for it; test is never read; blind is inference-only.

Training. One trainer covers all 139 runs: AdamW (weight decay 0.01), linear warmup over 6% of steps then linear decay, gradient clipping at 1.0, bf16, dynamic per-batch padding, max length 160, batch 32–64 (12–24 for large models), learning rate 1.5e-5–2.5e-5 (8e-6 for one divergence-prone large regressor), at most 8 epochs with early stopping on dev QWK (patience 2), dropout 0.1, seed 42, one GPU per job. Full per-family configurations, the tag glossary, and wall-clock costs are in Appendix B.

Evaluation. Local metrics mirror the official evaluator exactly. Because 18 free thresholds are fitted on a few thousand points, configurations carrying identical information still differ by a measured ±0.1–0.2 blind QWK, a noise floor below which no change shipped. Post-hoc comparisons pass through an *A/B firewall*: everything fitted (selection, weights, thresholds) is fitted on one half of a labeled split and scored on the untouched other half, over repeated permutations. Comparisons behind the official systems use the development split. The catalogs of rejected changes in Appendix F were instead measured on the public test split, the only split held out from training for every member family involved; no official system is fitted on it.

5 Results

5.1 Official Results

Table 1 reports the organizers' scoring of both official systems against the official baseline, a single fine-tuned AraBERTv2 `d3tok` regression model[2] and so the same encoder, text variant, and objective as one cell of Figure 4. On the ranked metric the systems clear the baseline by +2.5 (Strict)

[2] `CAMeL-Lab/readability-arabertv2-d3tok-reg`, linked from the official results page. Exact ZHR values from the organizers: Strict QWK 85.02, Open 85.17; we print one decimal, matching that page, which reports the baseline only at that precision.

System	QWK	Acc	±1	Dist	Acc_7	Acc_5	Acc_3
Baseline	82.5	**42.4**	**73.6**	1.1	**61.6**	**68.8**	75.6
ZHR Strict	**85.0**	38.8	72.0	1.1	60.4	68.7	**75.8**
Δ	+2.5	−3.6	−1.6	0.0	−1.2	−0.1	+0.2
ZHR Open	**85.2**	39.6	72.1	1.1	61.4	68.5	**75.8**
Δ	+2.7	−2.8	−1.5	0.0	−0.2	−0.3	+0.2

Table 1: Official blind-set results (organizers' scoring) against the official baseline (identical for both tracks). Δ = ZHR minus baseline; best per metric in bold. Acc = exact 19-level accuracy; ±1 = accuracy within one level; Dist = mean absolute distance (lower is better, tied here at the one-decimal precision of the results page); $\text{Acc}_{7/5/3}$ = accuracy after collapsing to 7/5/3 levels.

#	System	QWK	Acc	±1
Strict track (build order)				
1	first ensemble	84.1	33.3	70.4
2	6-member greedy, `d3tok` core	84.5	33.8	69.9
3	+1 member	84.4	34.8	71.4
4	+ objective diversity	84.6	35.9	73.2
5	**+ AraBERT-large = official (V6)**	**85.0**	**38.8**	**72.0**
Open track				
1	official Strict system, resubmitted	85.0	38.8	72.0
C	**official: clean 48, dev-fitted**	**85.2**	**39.6**	**72.1**

Table 2: Blind-set scores from the evaluation platform along each track's build path. Rows 5 and **C** are the official systems, as scored by the organizers (bold).

and +2.7 (Open) QWK; the baseline leads both on exact accuracy (+3.6/+2.8) and adjacent accuracy. Figure 2 shows why this is a trade-off rather than a defect of either system. QWK equals $1 - \text{MSE}/(\sigma^2_{\text{true}} + \sigma^2_{\text{pred}} + \Delta\mu^2)$, so it rewards predictions that hedge toward the center, while exact accuracy rewards committing to the mode. The two official systems differ by +0.2 QWK under an identical protocol, the measured joint worth of silver pretraining and clean-pool selection at the ensemble level.

5.2 The Build Path

Table 2 gives the build path of the official systems on the blind set. On Strict, objective diversity adds 0.1 QWK over the six-member `d3tok` core (0.2 over the seven-member step) and encoder capacity a further 0.4, giving the official system. Each step is a single blind observation near the ±0.1–0.2 noise floor (§4), so the trajectory, not any single step, carries the evidence. On Open, the official system's gain over the resubmitted Strict system (+0.2 QWK, +0.8 Acc) comes from silver pretraining and clean-pool selection under the identical dev-fitted protocol (§3.4).

Level	n	Recall (%)	Mean(pred−true)
1	44	86.4	+0.30
2	68	0.0	+1.25
3	182	51.6	+1.12
4	78	42.3	+1.18
5	417	13.9	+0.56
6	189	10.1	+1.32
7	701	55.9	+0.32
8	613	50.9	+0.55
9	236	45.8	+0.14
10	1,012	60.9	+0.33
11	409	23.2	−0.13
12	1,491	27.2	−0.49
13	349	37.5	−0.61
14	1,072	36.4	−0.31
15	258	43.4	−0.73
16	114	14.0	−0.97
17	49	26.5	−0.55
18	13	15.4	−2.08
19	15	53.3	−1.53

Table 3: Per-true-level dev diagnostics of the official Strict system. Easy vocalized levels are over-predicted upward; the modal level 12 and everything above it are compressed toward the center, the behaviour QWK rewards.

5.3 Error Analysis

Figure 3 (Appendix C) shows where the official Strict system errs on dev, the split the protocol assigns to error analysis; no blind labels exist. Errors are adjacent-heavy, but the costliest confusion cells concentrate on the modal level 12, which is systematically under-predicted toward the center: nine of the twelve largest QWK-numerator cells are true level 12, under-predicted by up to nine levels (recall 27.2%). The rarest levels also fail in order: the official system predicts level 2 only once across all of dev and never correctly (recall

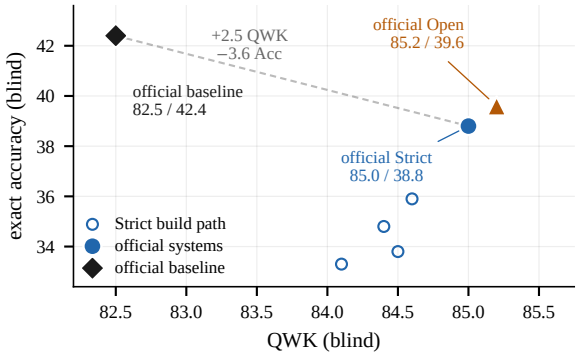


Figure 2: Exact accuracy vs. QWK on the blind set: the Strict build path (hollow), the two official systems (filled), and the official baseline, which sits 2.5 QWK below official Strict at 3.6 higher exact accuracy. The metrics reward different prediction styles.

0.0 on its 68 sentences; Appendix Table 3). The level-12 tail is systematic (all members are wrong together), which is why every variance-reduction change measures zero (Appendix F) and why the capacity step (row 5) was the last real modeling gain: it reduced this tail (blind Acc 35.9 $\rightarrow$ 38.8).

6 Conclusion

One recipe (encoders diversified by ordinal objective, greedy selection, dev-fitted threshold calibration) clears the official baseline by 2.5 QWK on Strict; with silver pretraining and contamination-aware selection it clears it by 2.7 on Open, while conceding exact accuracy (§5.1). Appendix F records what did not work.

Limitations

Protocol scope. During the evaluation phase we also submitted systems that trained on the development split and fitted thresholds on the test split. The organizers ruled these uses outside the task's data rules; they are not part of any official system and are not reported here.

Uncertainty. Blind deltas of $+0.1$–0.2 are single observations at display rounding, and we claim no statistical significance for them; our uncertainty statements are the A/B-firewall spreads and the ± 0.1–0.2 threshold-fit noise floor. Across the three seed-replica member pairs, solo dev QWK differs by -0.07 to $+0.29$, within that floor. The baseline comparison uses the published aggregate metrics; we did not run the released baseline model ourselves, so no cross-system significance test is possible. Training is seed-controlled but not bitwise deterministic.

Scope. The `d3tok` gain is measured on one controlled encoder pair and reverses on AraModernBERT; the silver gain ($+1.00$ and $+0.41$ on two matched pairs) is member-level, with an ensemble-level effect of $+0.10$–0.15; and the per-level error analysis is on dev, as no blind labels exist.

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A Method Details

Ordinal objectives. With levels $y \in \{1..19\}$ and logits z (width 1 for regression, 18 for CORN, 19 otherwise): *reg* minimizes $(\hat{y} - y)^2$ on the 1–19 scale; *CORN* (Shi et al., 2023) decomposes the label into 18 conditional binary tasks, task k trained on examples with $y \geq k$ to predict $y > k$, the continuous score being $1 + \sum_k \prod_{j \leq k} \sigma(z_j)$; *soft* (Díaz and Marathe, 2019) minimizes KL to the unimodal target $\mathrm{softmax}(-(l-y)^2/T)$, T=2; *wkl* (de la Torre et al., 2018) adds λ=0.5 times a differentiable QWK penalty with weights $(i-j)^2/18^2$ to cross-entropy; *emd* (Hou et al., 2016) minimizes $\|\mathrm{cumsum}(p) - \mathrm{cumsum}(\mathbf{1}_y)\|^2$. The latter three score by expectation $\sum_l l\, p(l)$.

Threshold calibration. A continuous ensemble score becomes a label via 18 ordered cut points, $\hat{y} = 1 + \sum_k [s > t_k]$. Cut points are fitted by cyclic coordinate ascent: initialized at the class midpoints, each t_k scans a grid (step 0.02, 8 rounds; a coarse 0.1/3 variant runs inside selection loops) between its neighbours, keeping a move only if QWK improves; monotonicity holds by construction. The Open track shrinks the fitted cuts a fraction k=0.35 of the way toward the label-prior quantiles of the score distribution, which keeps sparse levels populated.

Greedy selection. Forward selection with replacement (Caruana et al., 2004): at each round, add the member whose inclusion (weight = times picked) maximizes calibrated QWK of the running mean on the fit split; 25–30 rounds.

B Hyperparameters, Tags, and Engineering Record

All runs share: AdamW (weight decay 0.01, single parameter group), linear warmup over 6% of steps then linear decay to 0, gradient clipping at max-norm 1.0, bf16 autocast for training and inference (no gradient accumulation), dynamic per-batch padding (exact under mask-weighted mean pooling; 1.27–1.41× faster epochs), evaluation batch 64, early stopping on dev QWK under naive rounding with patience 2, dropout 0.1, and a single linear head (width 1 for regression, 18 for CORN, 19 otherwise). Per-family settings appear in Table 4; member-tag suffixes are decoded in Table 5. Each run used one GPU; epoch wall-clock in the committed training logs is 265–719 s (base–large).

Family	bs	lr	ep.	len
base encoders	32–64	2e-5–2.5e-5	5–8	160
`bert-large`	12–24	1.5e-5 / 8e-6[a]	4–6	160
silver stage (Open)	64	3e-5	1	160
XLM-R-large[b]	16	1e-5	–	160

Table 4: Training configurations: bs = batch size, lr = learning rate, ep. = max epochs, len = max tokens (seed 42 throughout; a seed-101 replica family exists). [a]8e-6 after a regression-head divergence in the large family. [b]Abandoned mid-training: ≈−2 naive QWK vs. base AraBERTv2 at 5× the epoch cost.

Tag	Meaning
`_d3` (or none)	`d3tok` text, train-only
`_raw`	raw text (normalized, diacritics kept)
`_d3n`	`d3tok` without rare-level upsampling (early A/B twin)
`_s2`	seed-101 replica of a `_d3` member
`_c2`, `_c3`	clean-protocol seed replicas (Open; seed 777)
`_slv`	silver-pretrained on BAREC-10M, then gold fine-tune
`_slvB`, `_slvlr1`	silver-stage configuration variants

Table 5: Member-tag glossary. A member name is *encoder_objective_tag*, e.g., `arabertv2_emd_slv`.

Three engineering findings are worth reuse. Cotenant training on one compute-bound GPU is a false economy: three concurrent jobs ran 4.2× slower per job, for zero net throughput. Caching every member's continuous scores made all post-training research GPU-free. And joint threshold calibration silently absorbs a member that collapses to constant output, so a member health check is worth running before trusting any ensemble's composition.

C Per-Level Error Analysis

Figure 3 shows the full dev confusion matrix of the official Strict system; Table 3 in §5.3 gives its per-level recall and mean signed error. The twelve costliest confusion cells by QWK-numerator share concentrate on true level 12 (nine of twelve, spanning predictions 3–14); the remaining three are true 11→7, 13→7, and 10→12. This systematic, all-members-wrong tail is what motivated the capacity step (§5.2) and what defeats every variance-reduction change in Appendix F. No blind labels exist, so error analysis is on dev.

D Silver-Pretraining Twin Comparisons

The Open track's member-level change is measured against identically configured, seed-matched

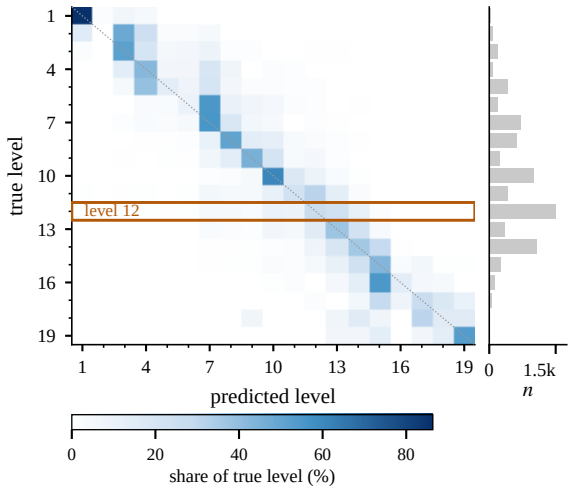


Figure 3: Official Strict system on dev: row-normalized confusion over the 19 levels, with per-level support n. The modal level 12 (orange) is systematically under-predicted toward the center; most of its mass lands on predicted levels 10–14 (§5.3).

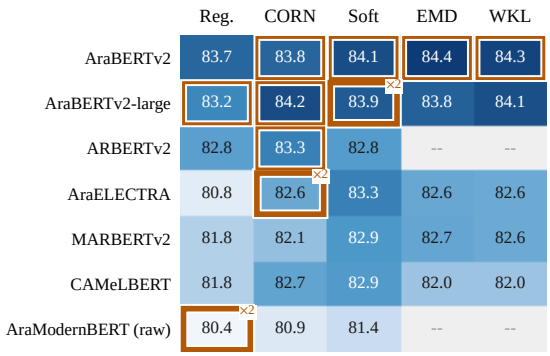


Figure 4: Solo dev QWK of every train-only backbone×objective member (thresholds fitted in-sample on dev; cells marked -- are members that were not trained). Orange outlines mark official Strict members, with ×2 tagging weight-2 members (the AraBERTv2/CORN slot holds its seed-101 replica); darker cells are higher. Selection does not follow solo strength: the weakest outlined member earns weight 2 by disagreeing usefully (§3.1).

twins: one epoch over the 438,996-sentence filtered BAREC-10M with automatic 19-level labels, then the identical gold fine-tune, with the 22,852 corpus rows matching a dev/test sentence removed first by normalized text match (Table 6).

Pair	Member	Solo QWK	Δ
emd	`arabertv2_emd_c2` (fresh)	84.93	
	`arabertv2_emd_slv` (silver)	85.93	+1.00
soft	`arabertv2_soft_c2` (fresh)	85.37	
	`arabertv2_soft_slv` (silver)	85.78	+0.41

Table 6: Silver-pretraining twins (Open). Solo QWK of members that were never selected or calibrated on the scoring split, computed once for this comparison with same-split cut points (in-sample convention, consistent within the column). After silver pretraining the gold stage peaks at epoch 0 (naive dev QWK 84.08 vs. 82.79 for the fresh twin) and declines thereafter. The ensemble-level gain is +0.10–0.15.

E Deployed Ensembles

Strict, official (V6, 10 members). Table 7: greedy selection on dev over the train-only pool, thresholds fitted on dev. The committed configuration is `configs/ensembles/v6.json` (members, weights, and cut points).

Open, official (18 weighted members). Greedy on dev over the full clean pool of 48. By weight: `arabertv2_large_emd_slv` and `arabertv2_soft_slv` 13.8% each; `arabertv2_emd_slv` and `aramodern_reg_raw` 10.3% each; `arabertv2_large_reg` 6.9%; and thirteen members at 3.4% each (`arabertv2_corn_s2`, `arabertv2_emd_d3`, `arabertv2_large_corn`, `arabertv2_large_emd`, `arabertv2_large_reg_raw`, `arabertv2_reg_s2`, `arabertv2_wkl_c2`, `arabertv2_wkl_slvB`, `araelectra_corn_d3`, `araelectra_soft_d3`, `aramodern_soft_raw`, `camelbert_reg_d3n`, `marbert_corn_d3`); cut points prior-shrunk at k=0.35 on dev.

Member	w
arabertv2_corn_s2	1
arabertv2_emd_d3	1
arabertv2_large_corn	1
arabertv2_large_reg	1
arabertv2_large_soft	2
arabertv2_soft_d3	1
arabertv2_wkl_d3	1
araelectra_corn_d3	2
aramodern_reg_raw	2
arbertv2_corn_d3	1

Table 7: The official Strict system: ten members and their greedy weights, selected and calibrated on dev.

Library composition. The Strict library holds 61 saved models over 7 Arabic encoders (objectives: 16 reg / 15 corn / 12 soft / 12 emd / 6 wkl; text: 55 d3tok / 6 raw); the 39 train-only members among them form the Strict selection pool. The Open pool holds the 48 clean members over 8 backbones, six of them silver-pretrained.

F Changes That Did Not Help

Negative results are reported here as first-class findings: one line per change. The Strict catalog is measured by repeated held-out A/B halves of the public test split, the Open catalog by cross-validation on the same split (§4); none of these changes was adopted, and the official systems are selected and calibrated on the development split alone. The Strict catalog has a unifying diagnosis: residual error is systematic, the large-error tail is all-members-wrong, and anything that assumes independent member noise therefore has nothing to average away.

Change	Outcome
seed replicas	$\sim$0; the level-12 tail is bias, not variance
dispersion rescaling	optimum $\alpha^* \approx 1.00$ already occupied
distribution matching	harmful without the true prior (to 77–78)
robust aggregation	median/trimmed at or below the mean; median's $\lvert e\rvert \geq 3$ rate worse
bagged thresholds	smooths across a sharp optimum
greedy over the full pool	overfits its selection half (-0.40 ± 0.17 over twelve dev half-splits)
ridge/simplex weights	ill-conditioned on correlated members
z-scoring / rank averaging	no gain
document-mean shrinkage	hurts at every $\lambda > 0$ despite 39% between-document variance

Table 8: Strict track: nine changes that did not help.

Change	Outcome
document context	$+0.006$; residual ICC 0.07
stacking (NNLS/ridge/GBM)	all below plain greedy
score normalization	-0.10; spreads differ only $1.2\times$
shift adaptation (affine/quantile)	-0.66 / -0.78 mean
spread rescaling γ	0 at $\gamma = 1$, negative elsewhere
uncertainty shrinkage	$+0.02$ best
corpus metadata features	residual R^2 0.002
pool pruning	strictly worse than the full pool
peer-model blend	0/4 analogues positive
label-free variance estimation	24.5% median error, biased

Table 9: Open track: ten changes that did not help (pool residuals correlate at $\sim$0.997, so combination-side changes measure flat).

Split	Use in the official systems
Strict	
train	training (54,845 sentences)
dev	early stopping, selection, cut points
test	never read
blind	inference only
Open (adds the BAREC-10M silver stage, 438,996)	
train	training, after the silver stage
dev	early stopping, selection, cut points
test	never read
blind	inference only; $\sim$57% of its sentences occur in the silver corpus (audited, §3.4)

Table 10: Data exposure per split for the official systems. Both tracks train on the training split (plus the silver corpus on Open) and use public checkpoints and CAMeL Tools; never SAMER, never blind labels.

G Complete Experiment Record

Table 11 indexes the experiments behind the official systems: the question each asked, what it fitted on which data, and its outcome. Each row corresponds to a committed experiment directory with its script, log, and result files, and the split roles use the vocabulary of Table 10.

H Reproducibility

Every reported number that can be recomputed is recomputed by an automated check: 61 strict-track and 51 open-track values, from cached member scores, committed configurations, and threshold sets; no GPU, corpus, or checkpoints are required. The chain closes on the submitted files themselves: both official predictions re-derive from the committed caches with 0 of 8,077 rows differing, and both submitted files were verified byte-identical (md5) against their committed copies. Their official scores are the organizers' scoring (2026-09-05), and the official Open system was built by a committed script (`exp019`) from the same caches. The camera-ready figures re-derive from the same records by one committed script. Analysis is fully deterministic (fixed seeds; split permutations derived from repetition indices); GPU training is seed-controlled but not bitwise reproducible. Software versions at training time were not logged (the environment record dates from documentation time), and the exact flags of one early member are unrecorded; both gaps are disclosed rather than reconstructed.

Record	Question	Trains / fits on	Outcome
Strict repository (extract; experiments/e01–e13)			
e01	which encoder×objective members work solo	train / dev	AraBERTv2-base family strongest; Figure 4
e03	does objective diversity beat encoder diversity	train / dev	yes: $+0.1$ blind (row 4 of Table 2)
e04	does encoder capacity help	train / dev	$+0.4$ blind (row 5); reduces the level-12 tail
e05	do seed replicas add ensemble value	train / dev	$\sim$0: the tail is bias, not variance
e06	greedy selection and weighting protocol	train / dev	the 10-member official system (V6)
e08	learned combiners vs. greedy	train / CV	stacking, ridge, GBM all below greedy
e09	pre-registered ship/no-ship gates	dev half-splits	the ship decisions behind Table 2
e12	post-processing changes over cached member scores	cached scores, held-out A/B	all within the noise floor; Appendix F
e13	dev-only re-selection gate	train / dev half-splits	wider dev-only re-selection loses 12/12 (-0.40 ± 0.17); V6 stands
Open repository (extract; full index in the repository)			
exp014	silver pretraining on BAREC-10M	silver+train / dev	$+1.00$ solo QWK vs. seed-matched twin (Appendix D)
exp010	does early stopping on a calibration split inflate its own scores	training logs	yes: best epoch $+1.06$ QWK over the mean epoch (n=18); motivates the clean pool
exp019	official Open system build	train(+silver) / dev	clean-48 greedy on dev; predicted $+0.24$ vs. V6, realized $+0.2$

Table 11: The experiment record behind the official systems. Every row re-derives from committed artifacts (Appendix H).